B. Covered Path

time limit per test 1 second
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 1400 | Tags: dp, greedy, math | https://codeforces.com/problemset/problem/534/B

The on-board computer on Polycarp's car measured that the car speed at the beginning of some section of the path equals v_1 meters per second, and in the end it is v_2 meters per second. We know that this section of the route took exactly t seconds to pass.

Assuming that at each of the seconds the speed is constant, and between seconds the speed can change at most by d meters per second in absolute value (i.e., the difference in the speed of any two adjacent seconds does not exceed d in absolute value), find the maximum possible length of the path section in meters.

Input

The first line contains two integers v_1 and v_2 ($1 \le v_1, v_2 \le 100$) — the speeds in meters per second at the beginning of the segment and at the end of the segment, respectively.

The second line contains two integers t ($2 \le t \le 100$) — the time when the car moves along the segment in seconds, d **($0 \le d \le 10$)** — the maximum value of the speed change between adjacent seconds.

It is guaranteed that there is a way to complete the segment so that:

- the speed in the first second equals v_1,
- the speed in the last second equals v_2,
- the absolute value of difference of speeds between any two adjacent seconds doesn't exceed d.

Output

Print the maximum possible length of the path segment in meters.

Examples

Input
5 6 4 2

Output
26

Input
10 10 10 0

Output
100

Note

In the first sample the sequence of speeds of Polycarpus' car can look as follows: 5, 7, 8, 6. Thus, the total path is 5 + 7 + 8 + 6 = 26 meters.

In the second sample, as $d = 0$, the car covers the whole segment at constant speed $v = 10$. In $t = 10$ seconds it covers the distance of 100 meters.